

B.Tech (Computer Science and Engineering) Semester-VII

Subject: Data Mining & Data Warehouse

Marks: 64

ASSIGNMENT #4

Section A (2 Marks Each)

[CO3] Q1. What are the data structures that are used for clustering?

Answer:

Common data structures used for clustering include:

1. **Array / Matrix (distance matrix)**
 2. **Graph-based structures (nearest neighbor graph)**
 3. **Tree-based structures (KD-tree, R-tree, CF-tree in BIRCH)**
 4. **Hash tables (for grid-based clustering)**
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[CO3] Q2. Discuss various issues regarding classification.

Answer:

Classification faces several issues such as:

1. **Overfitting & Underfitting**
 2. **Imbalanced datasets**
 3. **High dimensionality (curse of dimensionality)**
 4. **Noise & missing values in data**
 5. **Choosing suitable evaluation metrics**
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[CO3] Q3. Write the formula for arithmetic distance which is measured in K-means algorithm.

Answer:

K-means generally uses Euclidean distance, calculated as:

$$d(p, q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

Where p and q are data points in n -dimensional space.

[CO3] Q4. What does BIRCH stand for?

Answer:

BIRCH = Balanced Iterative Reducing and Clustering using Hierarchies.

[CO3] Q5. What is a dendrogram?

Answer:

A dendrogram is a **tree-like diagram** that shows how data points are merged or split step-by-step in **hierarchical clustering**. It visually represents cluster relationships and distances.

Section B (7 Marks Each)

[CO3] Q1. Write a short note on DBSCAN algorithm.

Answer:

DBSCAN (**Density-Based Spatial Clustering of Applications with Noise**) is a clustering algorithm that groups data points based on density. It forms clusters where points have a minimum number of neighbors within a given radius (**Eps**).

- **Key terms:** *Core point, Border point, Noise/outlier*
 - **Advantages:** Detects arbitrarily shaped clusters, identifies outliers automatically
 - **Limitations:** Sensitive to parameter choice, not suitable for varying density datasets
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[CO3] Q2. What are data points and their types?

Answer:

A **data point** is a single row or instance in a dataset representing an object with features.

Types of data points:

1. **Numerical (continuous, discrete)**

2. Categorical (nominal, ordinal)
 3. Binary (0/1, yes/no)
 4. Textual / Semantic data
 5. Time-series & spatial data
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[CO3] Q3. What is clustering and its types?

Answer:

Clustering is an **unsupervised learning technique** that groups similar data objects into clusters based on similarity/distance.

Types of clustering:

1. Partitioning (K-means, K-medoids)
 2. Hierarchical (Agglomerative & Divisive)
 3. Density-based (DBSCAN, OPTICS)
 4. Grid-based (STING, CLIQUE)
 5. Model-based (EM, SOM)
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Section C (11 Marks Each)

[CO3] Q1. Explain K-means algorithm with suitable example.

Answer:

K-means is a **partitioning clustering algorithm** that divides a dataset into **K clusters** based on similarity using distance measures (usually Euclidean distance). It is an **unsupervised learning** technique widely used in machine learning, pattern recognition, and data mining.

✓ Steps in K-Means Algorithm

1. Choose the number of clusters (K)
 2. Randomly select K points as initial centroids
 3. Assign each data point to the **closest centroid** (forming clusters)
 4. Recalculate new centroids by taking the **mean** of each cluster
 5. Repeat steps 3–4 until **centroids no longer change** (convergence)
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✓ Example Dataset

Point	X	Y
A	2	10
B	2	5
C	8	4
D	5	8
E	7	5
F	6	4

Let $K = 2$

Initial centroids selected: C(8,4) and A(2,10)

✓ Iteration 1 – Assign points to nearest centroid

Use Euclidean distance formula:

$$d(p, q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Cluster 1 (near A): A, B, D

Cluster 2 (near C): C, E, F

✓ Recompute centroids

Cluster 1 mean:

$$(2 + 2 + 5)/3 = 3,$$

$$(10 + 5 + 8)/3 = 7.67$$

Cluster 2 mean:

$$(8 + 7 + 6)/3 = 7,$$

$$(4 + 5 + 4)/3 = 4.33$$

New centroids $\rightarrow$ (3,7.67) and (7,4.33)

✓ Repeat assignment until stable (convergence)

Final clusters remain same $\rightarrow$ algorithm stops.

✓ Advantages

- ✓ Simple, fast, and efficient
- ✓ Works well on large datasets
- ✓ Easy to implement

✓ Limitations

- ✗ Requires predefined K
 - ✗ Sensitive to outliers
 - ✗ Works well only for spherical-shaped clusters
 - ✗ May converge to a local optimum
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[CO3] Q2. Differentiate between methods of hierarchical clustering.

Answer:

Hierarchical clustering builds a **tree-like structure (dendrogram)** to represent nested clustering. It is of **two types**:

Feature	Agglomerative (Bottom-Up)	Divisive (Top-Down)
Process	Starts with each point as a single cluster, then merges	Starts with one big cluster, then splits
Approach	Bottom → Up	Top → Down
Initial Step	N clusters → Merge stepwise to 1	1 cluster → Split stepwise to N
Complexity	$O(n^2 \log n)$	Higher than agglomerative
Usage	More common and practical	Less common, expensive
Example Algorithm	Single-link, Complete-link, Ward's method	DIANA
Dendrogram Direction	Small to large clusters	Large to small clusters

✓ Linkage Criteria (Used in Agglomerative)

1. **Single linkage:** min distance between clusters
2. **Complete linkage:** max distance between clusters

3. **Average linkage:** average pairwise distance
 4. **Ward's method:** minimizes total within-cluster variance
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✓ Advantages

- ✓ No need to pre-define number of clusters
- ✓ Good visualization using dendrogram
- ✓ Works on non-spherical clusters

✓ Disadvantages

- ✗ Once merged/split, cannot be undone
 - ✗ Computationally expensive for large datasets
 - ✗ Sensitive to noise and outliers
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[CO3] Q3. What do you mean by outlier analysis? Briefly describe types of outliers.

Answer:

Outlier analysis is a data mining task used to detect **data objects that deviate significantly** from normal patterns in a dataset. Outliers may represent **fraud, rare events, errors, or anomalies**.

✓ Why Outliers Occur?

- Human/system errors (wrong entry, sensor defect)
 - Rare valid events (fraudulent transaction, disease detection)
 - Natural extreme variations
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✓ Types of Outliers

Type	Description	Example
Global (Point) Outlier	A single data point very different from rest	Salary list: 30k, 35k, 40k, 9 lakh
Contextual (Conditional) Outlier	Outlier based on context (time, location)	Temperature 30°C normal in July, abnormal in December

Type	Description	Example
Collective Outlier	A group of data points that deviate together	Sudden 100 failed login attempts → cyber attack
Noise vs Outlier	Noise = random errors; outlier = meaningful deviation	Sensor glitch vs credit card fraud

✅ Detection Techniques

1. **Statistical methods** (Z-score, Boxplot, IQR)
2. **Distance-based methods** (k-NN outlier factor)
3. **Density-based methods** (LOF, DBSCAN)
4. **Clustering-based methods** (K-means – farthest point = outlier)
5. **Machine learning based models** (Isolation Forest, Autoencoders)

✅ Applications

- ✓ Fraud detection
 - ✓ Medical abnormality diagnosis
 - ✓ Network intrusion detection
 - ✓ Quality assurance in manufacturing
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